Project Overview

Dolaptan · MTM4681 Mobile Programming · Milestone · Team [TAKIM NO] · Hatice Karakuş, Muhammed Ali Demirsoy

1. Problem and user

Our users are students who share a home and small households in Türkiye. We expect them to face two connected problems. First, they open the fridge and cannot decide what to cook with what is there. Second, nobody knows exactly what is at home: food goes off, someone buys milk twice, and shopping is organized through a group chat. These are assumptions, and the interviews below test them.

Our answer: Dolaptan is a shared kitchen. Everyone in the home sees one pantry, gets recipes that use what is about to go off, and shares one shopping list. Cooking and shopping update the pantry, so it stays current without much typing.

2. Evidence

Interviews: [TO FILL: interview 5-10 people who cook at home, including people who share a flat. Write how many you spoke to, how many struggle to decide what to cook and how often, how they share shopping today (for example a group chat), what food went off at home last month, and one short quote. Do not submit this placeholder.]

Statistic: The UNEP Food Waste Index Report 2024 estimates 1.05 billion tonnes of food waste worldwide in 2022, and around 60% of it came from households. Its annex lists a Türkiye household estimate of 102 kg per person per year (about 8.7 million tonnes). This is a **low-confidence** value: the report found no household study for Türkiye, so the number is extrapolated from other countries. We use it as context, not as proof of our problem.

3. Why existing apps fall short

App	What we verified (2 Nov 2026)	Gap for our user
ChatGPT, Gemini	General assistants. They can suggest recipes from a photo	Not built to keep one structured pantry that several people update
FridgeBuddy	Shared pantry and list, expiry reminders, barcode scan. UK App Store listing: sharing through iCloud, household invites in Pro, meal ideas only with Apple Intelligence, Turkish not listed, 22 ratings	Paid household sharing, no Turkish, recipe ideas tied to Apple devices
the fifo	Free shared list and pantry with expiry countdowns (Product Hunt page)	No recipe suggestions mentioned
Nefis Yemek Tarifleri	Turkish recipe brand, 4.8 from 94k ratings on the Turkish App Store. The website has ingredient selection and an AI assistant. The app listing says it needs internet	No shared pantry or list. Photo detection not mentioned
SnapChef AI	Detects ingredients from a fridge photo. Premium $14.99 per month or $19.99-$29.99 per year	Single user, no shared pantry. Turkish [TO CHECK]

Pantry apps, recipe apps and AI assistants all exist. We did not confirm any app that combines a free shared household pantry, Turkish recipes that use what is about to go off, and a list that restocks the pantry. This is a desk check of store listings, not a full market review. **[TO CHECK: install FridgeBuddy and the fifo and test them before submission.]**

4. Must-have features and course topics

Must-have feature	Course topic
Add food by photo, then get 3 recipes that use expiring items (firebase_ai + App Check)	HTTP/API, JSON, async (Future)
Shared pantry and shopping list that update live, with Cooked and Bought rules	Firebase Auth, Firestore, Streams, StreamBuilder
Pantry copy and saved recipes that open without internet	Local DB (Hive)
Expiry reminders the day before	Device features: local notifications
Tabs, sheets, scan animation and recipe transitions	Navigation (go_router), animations
Separation of screens, state and data	MVVM, Provider

5. Weekly plan

Course week (topic)	Hatice Karakuş	Muhammed Ali Demirsoy
7 (milestone)	Interviews, Figma import, prototype links	Review PDFs, GitHub repository
8 (midterm)	Flutter project, go_router routes, screen skeletons	Firebase project, Auth, App Check debug setup
9 (Hive)	Pantry copy and saved recipes in Hive, offline banner	Camera, AI item detection, review screen
10 (Firestore, Auth)	Household, shared pantry and list streams, security rules	Recipe generation with JSON schema, Cooked and Bought loop
11 (Animations)	Scan animation, sheets, recipe transitions	Error states, retry, response validation
12 (Device, publishing)	Expiry reminders with local notifications	Two-account and real-phone tests, APK build, README
13 (Presentations)	5-minute demo and presentation	2-minute backup video, peer contribution form

6. Risks and responses

Risk	Response
The pantry gets out of date	Cooked and Bought update it automatically; states are only have, low and out, not exact counts
AI detects an item incorrectly	Users review every item, and uncertain ones are flagged before saving
AI suggests unsafe or allergenic food	Pass allergies in the prompt, show a reminder, and ask users to check
AI free-tier limit, outage or invalid JSON	Response schema, validation, one retry, then a retry message. The pantry and saved recipes still work
App Check blocks AI requests in the demo build	Firebase docs list required enforcement from 2 Nov 2026. Use the debug provider in development. For an APK outside Google Play, register its SHA-256, turn off PLAY_RECOGNIZED, and test early
Reminders do not fire	Ask for notification permission on Android 13+, and test on real phones; some Android brands stop background apps
Two people edit the same item	Last write wins; batched writes keep pantry and list consistent
Photos may contain personal content	Gemini free-tier input may be reviewed by people and used to improve Google products. Tell users to photograph only shelves, and do not store photos
Short schedule	Keep to the six must-haves. Cooking mode and barcode scanning wait until after the demo

Sources (accessed 2 Nov 2026)

UNEP (2024), Food Waste Index Report 2024: totals p. 64 of PDF, Türkiye data status p. 41, Annex 3 p. 190. Copy: sdg2advocacyhub.org/wp-content/uploads/2024/03/food_waste_index_report_2024.pdf

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